

Tax-Aware Portfolio Rebalancing and Tax-Loss Harvesting

A Lot-Level Simulation Engine and Strategy Playbook for Vise

UT Austin MS Business Analytics Capstone, Group 20

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Executive Overview

Vise builds and manages personalized portfolios for financial advisors, and its tax-loss harvesting (TLH) is a core commercial feature. The challenge is interpretability: advisors struggle to explain to clients why a particular harvest or rebalance occurred. Our team built a transparent, rules-based, lot-level simulation engine that quantifies the after-tax value of TLH across rebalancing strategies, portfolio constructions, and market environments, and produces an auditable trade log that an advisor can explain.

We backtested **192 matched TLH-versus-no-TLH comparisons** (384 individual simulations). The central finding is that harvesting works as a tax tool but its bottom-line value is decided by a second factor that is easy to overlook: **TLH reliably reduced taxes** (it cut cumulative tax in 86% of comparisons, by about +\$10,182 per \$1M on average), but to stay invested after selling at a loss the strategy must hold a wash-sale replacement ETF, and **the tracking gap between the original holding and its replacement usually moved portfolio value more than the tax saving did**. Net of everything, harvesting helped in only 55% of comparisons pre-liquidation and 37% after a full liquidation.

Business Problem and Objectives

Advisors leave platforms when they cannot explain automated decisions to clients, which raises service burden and compliance risk. The objective was not to replace Vise's optimizer but to build an explainable measurement tool: given a portfolio, a rebalancing rule, and a tax profile, quantify what TLH actually adds (or costs) after tax, and rank the strategies that deliver the best client outcomes in each market environment.

Business value

As an order-of-magnitude business case, a 10% reduction in advisor churn on a base of roughly 1,000 advisors retains about 10 advisors per year. At roughly \$25M of assets per advisor and a 30 bps fee, that is on the order of \$750K in retained annual revenue. The figure is an assumption-based estimate meant to frame the opportunity, not a measured result.

Data

The engine runs on a daily price history of roughly 700 ETFs and the corresponding dividend, stock split, and TLH proxy-substitution tables, all sourced from S&P Capital IQ and Vise. Prices feed a forward-filled, look-ahead-free price matrix; the proxy table maps each holding to ranked replacement ETFs for wash-sale-compliant harvesting. The full price extract ships with the project as a compact parquet so every result is reproducible from a clean checkout.

Engine Design

The simulation is lot-level. Every purchase creates a tax lot with its own cost basis and acquisition date, and every sale is classified as short-term or long-term using the IRS rule that a holding period of more than one year (366 or more calendar days) is long-term. The engine implements:

- **Tax accounting:** short-term and long-term netting, a \$3,000 ordinary-income offset, character- preserving loss carry-forward, and an annual settlement.
- **Wash-sale enforcement:** a 30-day lookback and 30-day forward block, with automatic substitution into a ranked proxy ETF so the position stays invested.
- **Rebalancing:** calendar (monthly, quarterly, yearly) and drift-band (absolute and relative thresholds), with costs embedded in NAV on every trade.
- **Costs:** a 12 bps round-trip charge (5 commission, 5 slippage, 2 bid-ask).

Methodology

Every cell of the grid was run through the same engine so the comparison is fair. The only difference between a matched pair is whether harvesting is enabled (a 10% loss threshold) or not; rebalancing dates, costs, and tax rules are identical. We hold \$1M of starting capital, 35% short-term and 20% long-term tax rates, and report results on a price basis so that calendar and threshold strategies, TLH-on and TLH-off, are all measured on the same footing.

Portfolios: a 40/60 and a 100/0 Target Allocation ETF model, a concentrated two-ETF mix (IVV/EFA), and a diversified three-ETF mix (SPY/TLT/GLD). **Periods:** the 2007-2008 bear, the 2010-2019 baseline, the 2023-2024 bull, and trailing 5, 10, and 20-year windows. **Strategies:** monthly, quarterly, and yearly rebalancing; absolute 5%, 10%, and 20% drift bands; and relative 25% and 50% bands.

For each matched pair we measure the **net after-tax value-add** of harvesting (the harvested portfolio's value minus the identical no-harvest portfolio's value) and decompose it into three drivers: **tax saved** (the reduction in cumulative tax paid), **replacement tracking** (the return difference from holding wash-sale substitute ETFs instead of the originals, plus the timing of any realized gains), and **extra cost** (the added transaction cost of harvest trades). Pre-liquidation value-add is measured while still invested; post-liquidation re-prices both portfolios after a full sale, which captures whether the benefit was permanent or merely deferral.

Validation

The pure-computation engine is covered by a unit-test suite with financially meaningful expected values: buy-and-hold parity, harvest firing and dollar-parity rebuys, wash-sale and proxy substitution, the \$3,000 offset and carry-forward, the short/long-term day-count boundary, dividend handling, and NAV reconciliation. Value-add is always computed against a baseline that rebalances on the same dates, so the comparison isolates the effect of harvesting alone.

Results

Net value-add by market period is positive in the short and medium windows and turns negative over the full 20-year window, where the replacement-tracking gap has the longest time to compound.

Market period	Net value-add (pre-liquidation)	Net value-add (post-liquidation)
Bear (2007-2008)	+\$3,558	+\$991
Baseline (2010-2019)	-\$11,617	-\$12,650
Bull (2023-2024)	+\$3,700	+\$1,721
Past 5 Years (2021-2025)	+\$12,243	-\$4,258
Past 10 Years (2016-2025)	+\$17,142	-\$3,494
Past 20 Years (2006-2025)	-\$15,353	-\$48,535

Decomposing the net value-add by portfolio shows what is really happening. **Tax saved is positive for every portfolio**, so as a tax tool TLH works. The net outcome, however, is decided by replacement tracking: it is strongly positive for the diversified 3-ETF and 2-ETF mixes and strongly negative for the two model-ETF portfolios. Tax was cut in 75% to 96% of runs, depending on the portfolio.

Portfolio	Tax saved	Replacement tracking	Extra cost	Net value-add	Avg TLH events
3-ETF (SPY/TLT/GLD)	+\$17,973	+\$15,371	-\$3,009	+\$30,335	19
2-ETF (IVV/EFA)	+\$8,860	+\$9,277	-\$2,259	+\$15,879	11
40/60 (TA ETF)	+\$8,524	-\$18,761	-\$3,038	-\$13,275	97
100/0 (TA ETF)	+\$5,370	-\$28,588	-\$3,272	-\$26,490	56

The reason is the replacement map. Several large-cap style holdings in the model portfolios (for example the S&P 500 Growth and Russell 1000 Growth sleeves) are replaced by a broad S&P 100 fund, which lagged the growth names by hundreds of percentage points over the long windows. After harvesting, the portfolio spends years holding the wrong index, and that drift, not the tax math, drives the large negative results. The single

worst run (100/0 (TA ETF), past 20 years, abs threshold 10%) was -\$492,226: replacement tracking was -\$410,475 while the tax effect was -\$73,185. The best run (3-ETF (SPY/TLT/GLD), past 20 years) was +\$210,784 pre and +\$137,351 post-liquidation, but it was tracking-dominated too: the replacement happened to outperform by +\$184,832, with a tax saving of only +\$31,924. Single-run extremes therefore reflect replacement tracking in both directions; the reliable, repeatable effect is the tax saving, positive in 86% of runs.

Strategy ranking

Ranking strategies by a composite of value-add, Sharpe, CAGR, drawdown, and information ratio, the clear pattern is that **less-frequent and wider-band rebalancing wins after tax**. Monthly rebalancing ranks worst overall and is worst in four of the six market periods.

Rebalancing strategy	Mean composite rank (lower is better)
Rel Threshold 25%	3.21
Abs Threshold 20%	3.25
Yearly Rebal	3.33
Abs Threshold 10%	3.50
Rel Threshold 50%	3.83
Abs Threshold 5%	4.67
Quarterly Rebal	5.62
Monthly Rebal	6.83

Recommendations

- **Replacement-ETF quality is the most important lever.** Map each holding to a closely tracking, style-matched substitute (growth to growth, value to value). The largest losses in this study came from replacing growth sleeves with a broad large-cap fund, not from the tax mechanics.
- **Treat TLH as a portfolio-design decision, not a default.** It adds value for diversified portfolios with good replacements; for concentrated or style-tilted portfolios, confirm the replacement tracks before enabling it.
- **Default to infrequent, wide-band rebalancing** (yearly, relative-25%, absolute-20%); monthly and quarterly rebalancing churn the portfolio and add cost without improving the after-tax result.
- **Lead client communication with post-liquidation figures**, since much of the in-flight tax benefit is deferral that shrinks once gains are realized.

Execution Plan

We propose a phased, roughly nine-month rollout: curate and validate the replacement-ETF map against realized tracking, integrate the measurement engine with live model portfolios, run a pilot with a 15 to 25 advisor cohort, instrument the auditable trade log inside the advisor tooling, and track a target reduction in TLH-related service tickets before a broader release.

Limitations

- Net value-add is reported on a price basis for cross-strategy comparability; the interactive app can additionally model dividends and reinvestment.
- Results depend heavily on the supplied replacement-ETF map; a different (better-tracking) map would materially improve the harvested portfolios.
- The Sharpe ratio is defined as CAGR over annualized volatility at a 0% risk-free rate, which is internally consistent for ranking but differs from the textbook definition.
- The price universe covers active tickers, so single-security history carries some survivorship bias.
- Dividend timing uses payment date rather than ex-date, and only federal short and long-term capital gains plus the ordinary-income offset are modeled (no state or local taxes). Portfolios are long-only.

Conclusion

Across 192 matched comparisons, tax-loss harvesting is a reliable tax tool whose bottom-line value is gated by the quality of the wash-sale replacement. Harvesting cut taxes in the large majority of runs, but because it forces the portfolio to hold substitute ETFs, the net result hinges on how closely those substitutes track. The deliverable is not a blanket "always harvest" rule but an explainable engine, a clear decomposition of where harvesting helps and hurts, and a ranked playbook an advisor can defend.

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